

Assignment 3: Distance Unit Conversion GUI Application

Assignment for COSC 1100 – Intro to Programming

Prior to attempting this problem, you should have done the following:

1. Viewed the User Interfaces in Python (tkinter) videos and other content from week 10-11 and completed the associated Pre-Class Activity.

General Requirements

1. This assignment is to be completed **individually**. Group submissions will not be considered.
2. Analyze the problem, design a documented plan (i.e. form sketch, widget properties plan, flowchart or pseudo-code), design and implement your user-interface based application **following the step-by-step approach** presented in this course.
3. Submit your solution and plan to the appropriate assignment folder on DC Connect **by the due date provided**.
4. Your instructor will assign a grade and post feedback on your submission to DC Connect.

Program Requirements

For this assignment you will have to plan, create and implement a user interface using Python with Tkinter. The user interface is a window that will accept either a distance value as either kilometres (km) or miles (mi) and convert that measurement to the other scale. The user will select whether they are converting from kilometres to miles or from miles to kilometres.

The user should be allowed to enter a distance in a text box, select one of the two distance units to convert the value to and click on a button to cause the conversion and results to be displayed in a label on the form. These distance conversions will only be completed on valid numeric distances; if the entry is not numeric, an error message should be displayed to the user when they attempt the conversion.

Additionally, the user should have an option for clicking a button to clear the form of all of its entries and results (or error messaging), and ultimately setting the form to its default state for new entries. The user should also have the capability to click a button that would close the window and end the program.

Detailed Requirements for the User Interface:

- Use the resources provided to plan and build the user interface for this assignment as follows. Pay keen attention to all the required naming conventions as outlined in the style guide used in this course.
- Utilize the grid geometry manager to place widgets in specific rows and columns; your plan should reflect this too
- Include one entry textbox with an associated label for entry of the value to convert.
- Include two radio buttons for selection of distance units to convert to, “Kilometres” or “Miles”. When the program starts, or after a Clear operation the default selection should be “Kilometres”.
- Include a label to display the results of the conversion, or an error message on a blank or invalid entry, this label should not have any other message visible when the program starts.
- Include three buttons; one to initiate the conversion of the entered value to the selected distance unit scale; one to clear the text box and the results label as well as set the scale back to the default (Kilometres) and set the focus in the entry text box; and a third button to exit the program.
- Ensure your design has a tidy, professional appearance similar to Windows standards and is keyboard friendly. Be sure to include tooltips by some means, for example, by using Balloons.

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Detailed Requirements for Functionality:

- When the user clicks the button to convert, the first step is to validate the user input in the entry textbox. Any real number is acceptable. If the user input in the textbox is not a real number, display an appropriate message in the results output label as follows.
“ERROR: Please enter a numeric distance to convert.”
- If the user input is a real number, store it, then calculate the converted distance value based on which radio button is selected. This output should be rounded to two decimal places, and displayed in the output label. Follow this example for the appropriate format:
“60.2km converts to 37.41mi.”
- When the user clicks the button to clear, the window should be reset to its default state, ready for an entry. **Hint: there may be things to do other than just clearing both the input entry textbox and the results output label.**
- When the user clicks the button to exit, the window should close, ending the program.

Accessibility Considerations:

To be eligible for full marks on this assignment, your application needs to meet a certain standard of accessibility. For our purposes, this means the following:

- The Enter key, when pressed, should attempt to activate the button that initiates the conversion.
- There should be a keyboard combination (e.g. “Alt-c”, “Alt-x” ...) to access each of the three buttons.
- The order in which controls are selected when the tab key is pressed should be sensible and convenient.
- The colours used on the window and widgets should be high-contrast and text should be easy to read.
- Valid tooltips or other descriptive help text must be available by some means; the best option is likely to use the Tix “Balloon” class as described in the video series related to this lab, although an updating status bar is another option.

If the application cannot be used easily without the use of a mouse or trackpad, you have not succeeded with your GUI design.

Style Guide:

To be eligible for full marks on this or any assignment in this course your application must conform to the requirements as outlined above as well as our prescribed style guide, in this case making sure to observe the PEP8 naming conventions for Python as well as appropriate and complete program documentation.

Development Hints:

Your focus for this assignment should be the effective planning and design of a user interface that is intuitive and uncomplicated by extra, unnecessary controls.